# AYAZ MEHMOOD

AI Researcher — Medical Imaging, Generative Modelling, Multimodal Learning

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## ABOUT ME

AI researcher working on trustworthy medical imaging and generative modelling. My work spans domain shift and harmonisation across acquisition devices, synthetic data generation with diffusion and flow-matching models including virtual stain translation, prototype-based explainability, multimodal and vision-language modelling, and longitudinal clinical risk prediction on electronic health records. One manuscript is under review, a second is complete and ready for submission, and a journal article is submitted. I hold an MS in Data Science and graduated first in my class in Computer Systems Engineering. I am seeking a fully funded PhD position in these areas.

## RESEARCH EXPERIENCE BY THEME

**Domain shift, harmonisation and robustness**

- Identified performance degradation across acquisition devices and tissue density through stratified analysis and external validation, where aggregate metrics concealed the drop.
- Implemented device unlearning to quantify vendor-specific information retained in learned representations.
- Developing cross-device harmonisation of mammograms prior to classification; preliminary models improve on the unharmonised baseline (ongoing).

**Synthetic data generation and virtual staining**

- Developing H&E to IHC virtual stain translation using flow matching; current results exceed the baseline (ongoing).
- Earlier approach to lesion synthesis on VinDr-Mammo used LoRA-adapted latent diffusion inpainting; it did not improve downstream performance, as natural-image priors favour seamless blending over diagnostically faithful structure. This motivated the move to flow-matching formulations with frequency-domain constraints.
- Generative modelling beyond medicine: Flux text-to-image fine-tuning, Stable Diffusion virtual try-on and try-off, Pix2Pix GAN face unmasking.

**Multimodal learning and vision-language models**

- Fine-tuned BLIP and CLIP for medical report generation from tuberculosis chest imaging.
- Machine unlearning in large language models via gradient ascent with KL-divergence regularisation (SemEval-2025 shared task).
- Medical report summarisation with large language models; information extraction from doctor–patient conversations using retrieval-augmented generation.
- Parameter-efficient fine-tuning of quantized LLMs (Llama 2, Llama 3, Mistral, Falcon) for Urdu summarisation.

**dExplainable AI**

- Finding-BI-RADS prototype learning for mammogram classification, grounding predictions in defined radiological categories rather than post-hoc saliency (manuscript under review).
- Applied explainable AI to EMR and longitudinal clinical models at Klarity, covering cancer risk, chronic disease and mortality prediction, to make outputs interpretable for clinicians.

**Longitudinal and clinical data modelling**

- Five-year cancer risk prediction from demographic and clinical data on longitudinal EHR (manuscript submitted to Cancer Epidemiology).
- Chronic disease progression, biological age (BioAge) and mortality models on linked EHR and insurance records with irregular follow-up.

**Medical imaging (additional)**

- Early detection of Alzheimer's disease using Vision Transformers; tuberculosis classification and detection (ViT, YOLOv5); skin cancer segmentation (U-Net, DeepLabV3, TransViT); arrhythmia detection from ECG signals.

## WORK EXPERIENCE

**[ 10/2024 – Current ] Data Scientist** – Managed Self Limited (Klarity)
*Islamabad, Pakistan*

- Built machine learning models for five-year risk of common cancers from demographic and clinical data; co-author of the manuscript submitted to Cancer Epidemiology.
- Developed chronic disease progression, biological age and mortality models on longitudinal EHR and insurance data.
- Applied explainable AI across EMR and longitudinal clinical data to make model outputs interpretable for clinicians.
- Collaborated with cross-functional teams to integrate validated models into production healthcare products.

**[ 10/2022 – 07/2023 ] Research Assistant** – National Center of Artificial Intelligence (NCAI), Medical Imaging and Diagnostic Lab
*Islamabad, Pakistan*

- Member of the team that developed the laboratory's first AI-based DICOM viewer for medical imaging.
- Developed classification, detection and segmentation models for breast cancer, tuberculosis and brain tumour imaging, in collaboration with radiologists.

**[ 07/2023 – 11/2024 ] Data Science and Generative AI Engineer** – Glowlogix; Dream Shake Media LLC (part-time)
*Islamabad, Pakistan; Remote*

- End-to-end resume parser, employee-retention and applicant-suitability models, face recognition backed by a vector database.
- AI backend and Flux-based custom logo generation for creativecor.com.

**[ 04/2022 – 10/2022 ] AI Research Scholar** – National Center for Cyber Security (NCCS), Smart IoT Lab
*Islamabad, Pakistan*

- Designed and tested AI systems for industrial applications; built a real-time face-recognition attendance system.

## EDUCATION AND TRAINING

**[ 2023 – 2026 ] Master of Science in Data Science** – National University of Sciences and Technology (NUST)
*Islamabad, Pakistan | EQF level 7 | Final grade: CGPA 3.60 / 4.00*

- Thesis: breast cancer classification from dual-view mammography using multi-view fusion, Finding-BI-RADS prototype learning and LoRA-adapted foundation models.

**[ 2018 – 2022 ] Bachelor of Science in Computer Systems Engineering** – University of Engineering and Technology (UET) Peshawar
*Peshawar, Pakistan | EQF level 6 | Final grade: CGPA 3.74 / 4.00 — ranked first in class*

- Final-year project: anomaly detection from video streams using deep learning techniques.

## PUBLICATIONS

**Under review**

- “Intelligent Breast Cropping and Multi-View Fusion with Finding-BI-RADS Prototype Learning for Mammogram Classification”, IEEE Access (under review).

**Submitted**

- “Machine Learning-Enabled Pre-Screening for Five-Year Risk of Common Cancers Using Demographic and Clinical Data”, Cancer Epidemiology (submitted, manuscript no. CANEP-D-26-01959).

**Completed, ready for submission**

- “Adaptive LoRA Foundation Model Learning for Breast-Level Classification from Dual-View Mammography”.

**Preprints and shared tasks**

- “A Named Entity Recognition and Topic Modeling-Based Solution for Locating and Better Assessment of Natural Disasters in Social Media”, arXiv:2405.00903, 2024 — cited 23 times.
- “Unlearning from LLM using Gradient Ascent plus KL Divergence”, SemEval-2025 shared task.

## DIGITAL SKILLS

- **Imaging and domain adaptation:** DICOM, multi-view fusion, segmentation, detection, harmonisation, unlearning, stratified and external validation.
- **Generative modelling:** diffusion models, flow matching, virtual stain translation, Stable Diffusion, Flux, GANs, inpainting.
- **Foundation models and VLMs:** Vision Transformers, BLIP, CLIP, LoRA and PEFT, quantization, LLMs (Llama, Mistral, Falcon), RAG, LangChain, function calling.
- **Explainability:** prototype learning, attention and saliency methods, calibration analysis.
- **Clinical data:** EHR/EMR, longitudinal and irregular follow-up data, insurance data, survival and risk modelling, ECG signals.
- **Engineering:** Python, PyTorch, SQL, Git, Docker, Linux, HPC/GPU clusters, Flask, AWS, SageMaker.

## HONOURS AND AWARDS

- Ranked first in class, BSc Computer Systems Engineering, UET Peshawar (2022).

## REFERENCES

**Dr. M. Daud Abdullah Asif** — Master's Thesis Supervisor, School of Electrical Engineering and Computer Science (SEECS), NUST, Islamabad, Pakistan
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